

M5 Sales Data

EDA & Feature-Design Report

What the dataset contains, what patterns exist in demand, and what a future forecasting model might use

Item	Value
Source dataset	processed_dataset/sales_long_full.parquet
Total rows analyzed	59,181,090
Series / items / stores	30,490 / 3,049 / 10
Date range	2011-01-29 to 2016-05-22 (1,941 days)
Zero-sales rows	40,241,819 (68.0%)
Model trained	No — EDA and feature-design only
Novelty decided	No — candidate directions only, for team discussion

1. Objective

Answer three questions before the team commits to a modeling strategy: **What does the dataset actually contain? What patterns and problems exist in demand? What information should we consider giving a future forecasting model?** Every claim in this report is backed by a statistic, table, or chart generated directly from the data — see EDA_METHODOLOGY.md for exactly how, and the EDA/statistics, EDA/tables, and EDA/charts folders for the underlying evidence.

2. Dataset Used

Source: processed_dataset/sales_long_full.parquet — the validated base dataset built and documented in processed_dataset/PROCESSING_REPORT.md. This EDA reads that file only; it never touches raw_dataset/ and never writes to the Parquet file itself.

3. Dataset Structure (Phase 1)

Metric	Value
Total rows	59,181,090
Total columns	22
Unique series	30,490
Unique items	3,049
Unique stores	10
Unique departments	7
Unique categories	3
Unique states	3
Date range	2011-01-29 to 2016-05-22
Number of days	1,941
Duplicate (id, date) pairs	0
Missing sell_price	12,299,413 (20.78%)
Sales min / max / mean / median	0 / 763 / 1.131 / 0.0
Total sales (units)	66,927,173
Zero-sales rows	40,241,819 (68.00%)
Non-zero sales rows	18,939,271 (32.00%)

Full detail: EDA/statistics/phase1_sanity_check.json. These figures match processed_dataset/PROCESSING_REPORT.md exactly.

4. EDA Methodology

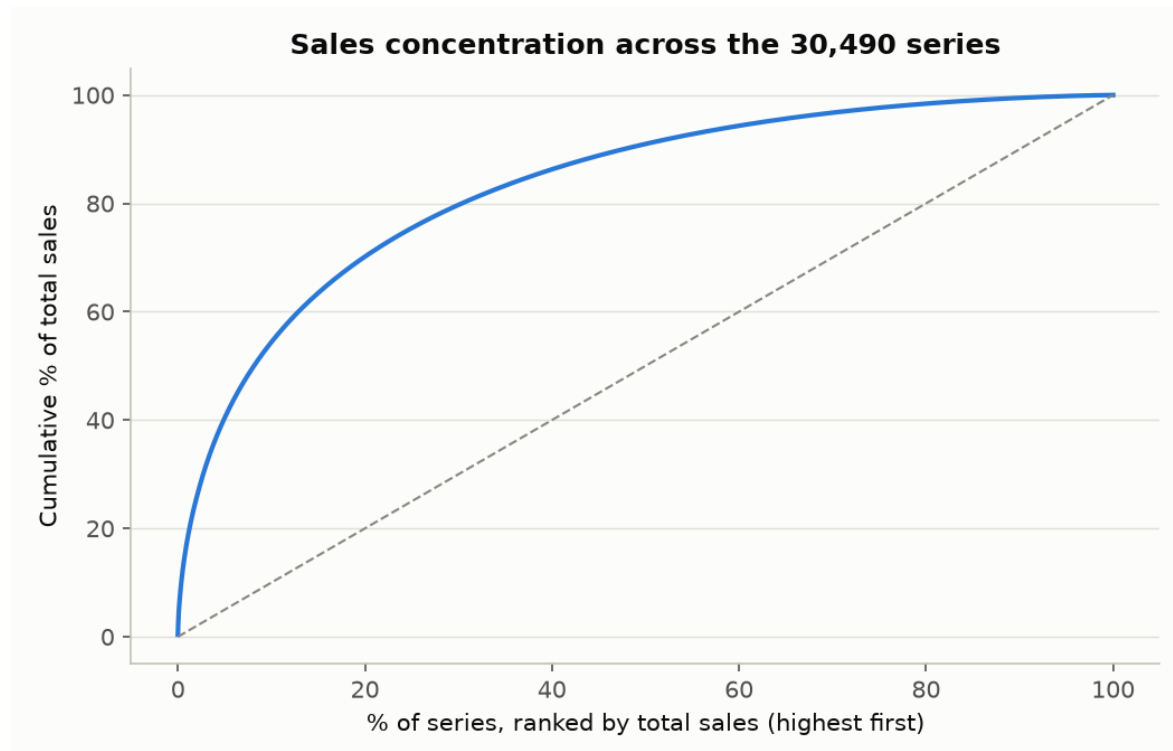
Full detail in EDA_METHODODOLOGY.md. In short: every phase runs directly against the full 59.18M-row table (one exception below), time-order-dependent computations explicitly sort by (id, date) first (the file's physical storage order is date-major, not id-major), zero values are never altered, and activity-class thresholds are derived from this dataset's own distribution rather than picked in advance.

The one sampling exception: Phase 8's per-series event/price sensitivity uses a stratified random sample (seed=99, n=501 of 30,490 series, proportional across category x volume-tier). Everything else — including all lag/rolling correlations and the zero-streak analysis — runs on the complete dataset.

5. Sales Distribution Findings (Phase 2)

What/why: the shape of the sales value distribution, mean vs. median, percentiles, and how total sales spread across categories/departments/stores/states/series — needed because a forecasting approach must match the actual shape of the target.

- The distribution is extremely right-skewed: skewness = 17.1 across all rows (11.9 even after dropping zeros).
- Mean (1.131) is far above the median (0.0) — a classic zero-inflated, long-tailed count variable.
- Top 10% of series by total sales account for 54.4% of all units sold; bottom 10% account for only 0.52%.
- FOODS accounts for 68.6% of total sales despite being 1 of 3 categories; FOODS_3 alone is 49.2% of company-wide sales.
- 51.4% of series have a coefficient of variation above 2 — more than half the catalog is highly volatile relative to its own average.



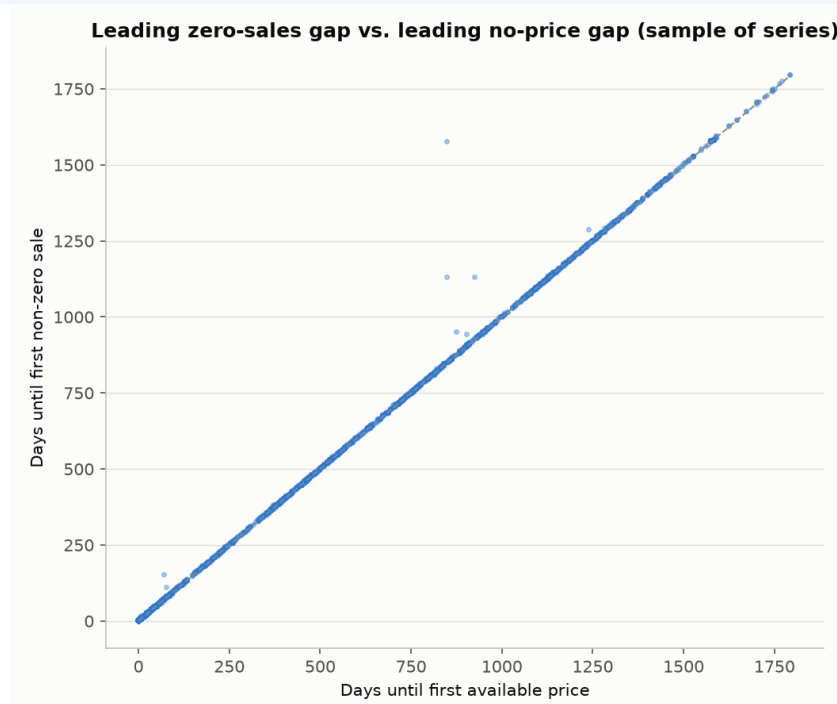
Charts: 01-05. Tables: sales_by_category.csv, sales_by_department.csv, sales_by_store.csv, sales_by_state.csv, series_level_sales_stats.csv.

6. Zero-Sales / Intermittent Demand Findings (Phase 3) — HIGH PRIORITY

What/why: with 68% of rows at zero, how those zeros are distributed and structured matters more for modeling strategy than almost anything else in this dataset.

- Zero% varies by category (HOBBIES 77.1% vs. FOODS 61.8%) and store (CA_3 59.4% vs. CA_4 72.0%).
- Per-series zero% spans a wide distribution (p5=24.7%, p50=73.3%, p95=95.0%). Data-driven classes: 7,629 High-activity, 15,267 Regular/intermittent, 6,103 Sparse, 1,491 Extremely sparse.
- Zero-run lengths: median 2 days, mean 6.08 days, p99=52 days, max=1845 days.
- KEY FINDING: leading zero-sales gaps align almost exactly with leading no-price gaps — median difference of only 3 days. and 99.48% of series are within 7 days of each other.

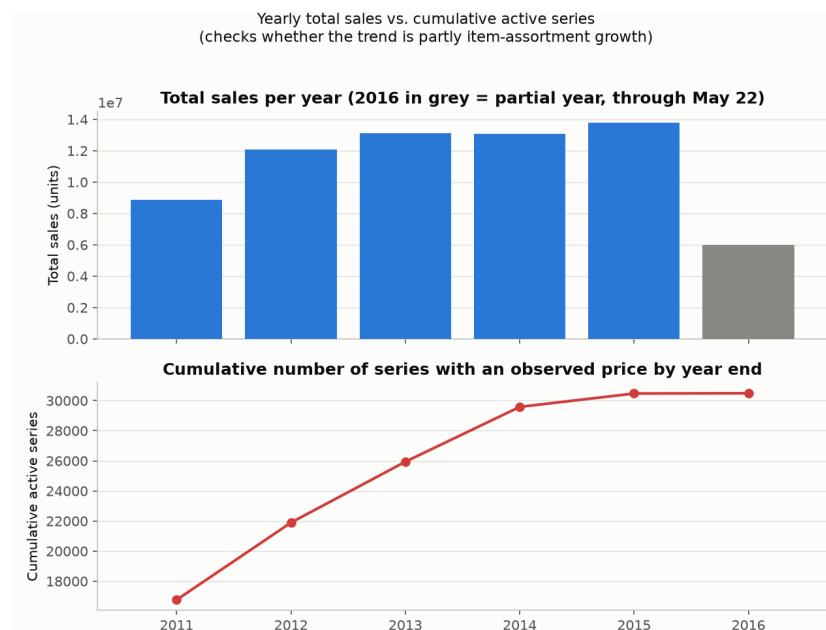
FACT: leading zero-sales periods and leading no-price periods end at almost the same time for the vast majority of series. INTERPRETATION (hypothesis, well-supported): consistent with items simply not yet being listed/available early on, rather than listed-but-not-selling. This is NOT a stockout claim — the dataset has no inventory field.



Charts: 06-09. Tables: zero_pct_by_*.csv, series_zero_pct.csv, zero_streak_*.csv, series_leading_zero_and_price_gap.csv.

7. Time / Seasonality Findings (Phase 4)

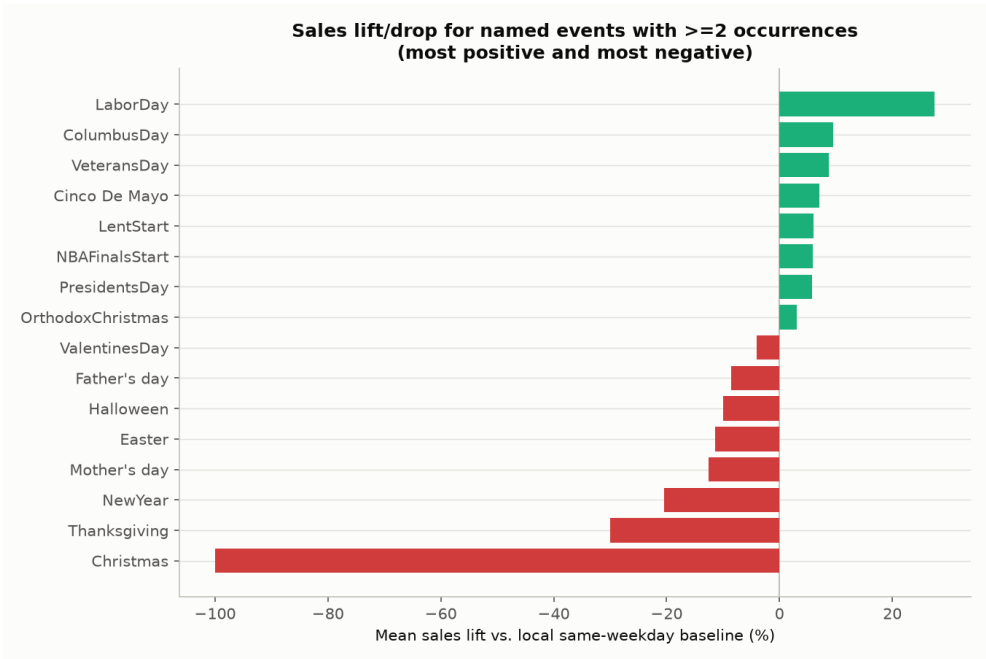
- Weekends are meaningfully higher: mean sales on Sat/Sun (1.361) are 31.1% higher than weekday-only mean (1.038), holding within every category.
- Monthly effects are real but modest: highest month (#8) vs lowest (#12) is a 8.99% spread.
- The apparent yearly sales growth (2011-2015) is at least partly a composition effect: active-series count nearly doubled (16,762 to 30,474) over the same window the trend appears in.
- 2016 is a partial year (data ends 2016-05-22) — its lower total is a coverage artifact, not declining demand.



Charts: 10-14. Tables: mean_sales_by_weekday_and_category.csv, mean_sales_by_month_and_category.csv, yearly_sales_stats.csv, daily_total_sales.csv.

8. Event / Holiday Findings (Phase 5)

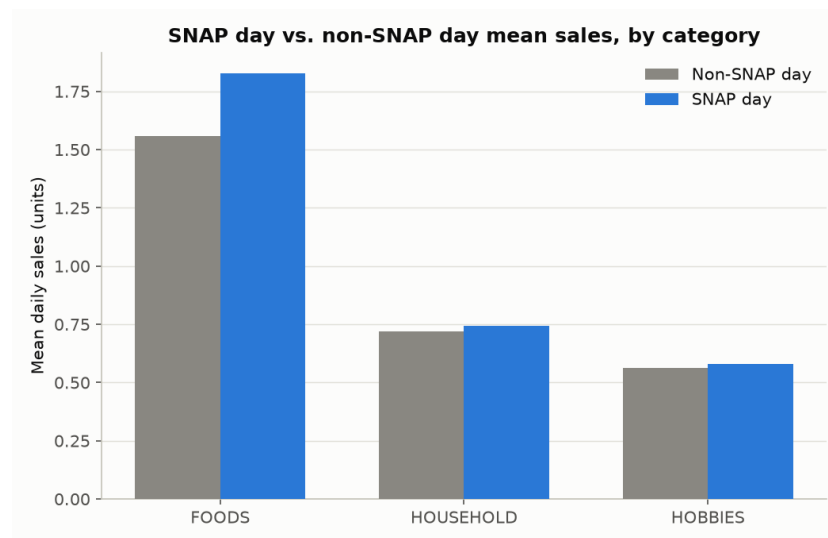
- In aggregate, event days look unremarkable (-4.6% vs non-event) — but this masks large, opposite individual effects.
- Christmas: -99.95% vs local baseline (stores essentially closed). Thanksgiving: -29.9%. NewYear: -20.4%. Mother's day: -12.5%.
- LaborDay: +27.5%. ColumbusDay: +9.6%. VeteransDay: +8.8%.
- A single 'is_event' flag would badly underuse this signal — the specific event identity carries the useful information.



Charts: 15-16. Tables: event_effect_by_*.csv, event_name_aggregated_lift.csv.

9. SNAP Findings (Phase 6)

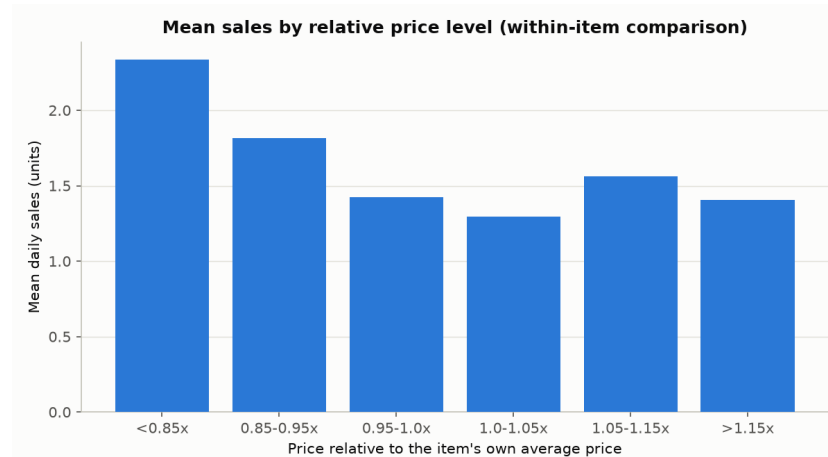
- SNAP days show 12.7% higher mean sales overall than non-SNAP days.
- Effect concentrates in food: FOODS +17.3% vs HOUSEHOLD +3.5% and HOBBIES +2.5%. FOODS_2 shows +32.3% — the largest of any department.
- Effect holds directionally in all 3 states but varies: WI +21.8%, TX +11.5%, CA +8.0%.
- This is one of the more trustworthy signals precisely because the effect lands where domain knowledge predicts (food spending) — a genuine internal consistency check.



Charts: 17-18. Tables: snap_effect_by_*.csv.

10. Price Findings (Phase 7)

- Price changes are infrequent: 78,057 change events across the dataset, mean 2.56 changes per series over ~5 years (0.159% daily change rate).
- Within-item relative price shows a mostly sensible pattern (lower relative price -> higher sales) but is NOT perfectly monotonic — an unexplained uptick appears at 1.05-1.15x.
- SURPRISING: sales rose after BOTH price increases (+71.0% mean) and decreases (+48.5% mean), while the MEDIAN change in both directions was 0% — suggesting price changes often coincide with other demand-moving events rather than causing the lift.
- This is a HYPOTHESIS, not a confirmed effect — there is no promotion field to confirm causality.

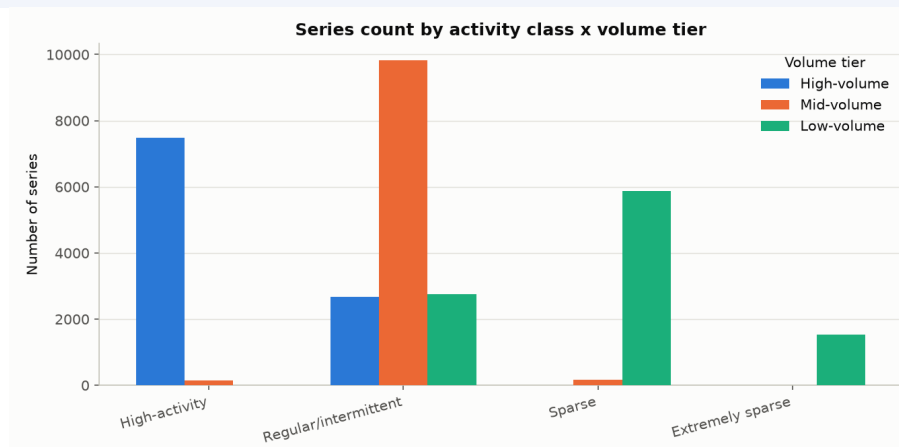


Charts: 19-22. Tables: price_distribution_by_category.csv, sales_by_relative_price_bin*.csv, price_change_event_sales_before_after.csv.

11. Product / Store Heterogeneity (Phase 8)

- Crossing activity class with volume tier shows real segments, not a smooth continuum.
- Per-series seasonality strength varies widely: CV of monthly means ranges from 0.136 (p10) to 0.557 (p90) — a >4x spread.
- Sampled event sensitivity (n=501): 63.3% of series show LOWER sales on event days.
- Sampled price-sales correlation: about 65% negative, 35% positive among series with enough price variation to measure — majority-negative but far from unanimous.

CONCLUSION supported by this evidence: behavior is genuinely heterogeneous across category, department, store, activity level, seasonality, and (more weakly) price/event sensitivity. A single one-size-fits-all forecasting approach is unlikely to perform equally well across all segments — a data-supported statement, not an assumption.

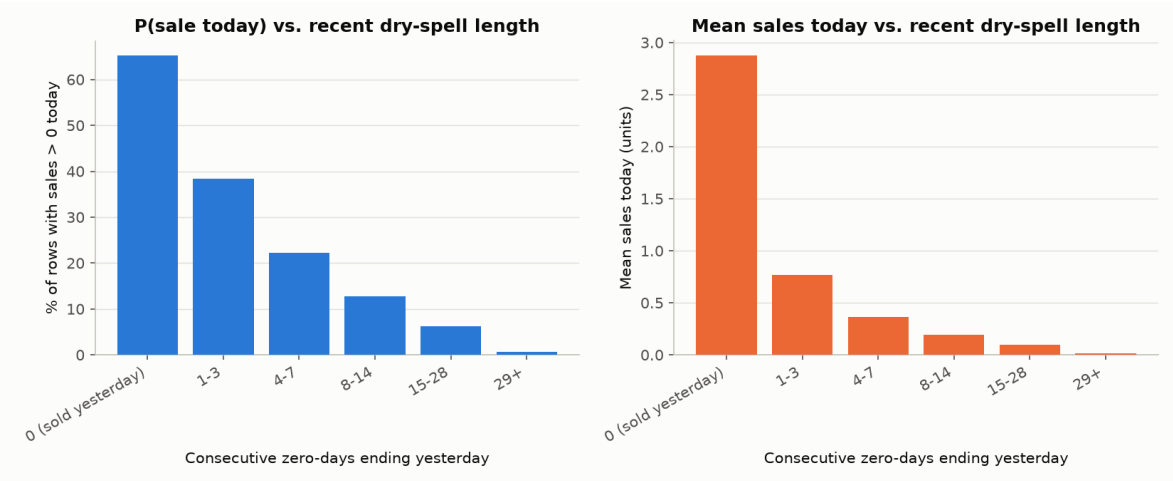


Charts: 23, 26. Tables: segment_activity_x_volume.csv, series_seasonality_index.csv, sampled_series_event_price_sensitivity.csv.

12. Important Relationships (Phase 9)

Relationship	Correlation
lag_1 (yesterday)	$r = 0.768$
lag_7	$r = 0.720$
lag_14	$r = 0.689$
lag_28	$r = 0.672$
rolling_mean_7 (prior 7 days)	$r = 0.820$ (strongest found)
rolling_mean_28 (prior 28 days)	$r = 0.807$
sell_price	$r = -0.151$
is_snap_day	$r = 0.017$
is_event_day	$r = -0.004$

The single cleanest relationship in the whole EDA: P(sale today) falls from 65.2% (sold yesterday) to 38.3% (1-3 days dry) to 22.2% (4-7) to 12.7% (8-14) to 6.2% (15-28) to just 0.6% after 29+ consecutive zero days — a near-perfect staircase.



Charts: 24-25. Table: days_since_last_sale_vs_today.csv.

13. Candidate Feature Engineering (Phase 10)

27 candidate features were evaluated, each grounded in a specific finding above. Full detail (including data source, evidence, and leakage risk for every feature) is in EDA/tables/feature_candidates.csv. Highest-priority features below.

Feature	Priority	Key evidence
days_since_last_sale (as of origin)	High	P(sale) 65.2% -> 0.6% across dry-spell buckets (Phase 9)
rolling_mean_7 (as of origin)	High	r=0.820 with same-day sales - strongest found (Phase 9)
rolling_mean_28 (as of origin)	High	r=0.807 with same-day sales (Phase 9)
lag_28 (relative to origin)	High	Only single lag safe for all 28 horizon days (Phase 9/11)
event_name	High	Christmas -99.95% vs LaborDay +27.5% (Phase 5)
snap indicator (state-matched)	High	+12.7% overall, +17.3% FOODS, +32.3% FOODS_2 (Phase 6)
days_since_first_listing	High	99.48% of series: leading-zero gap within 7 days of leading-no-price gap (Phase 3)
category/dept/store/state hierarchy	High	Substantial behavioral differences at every level (Phases 2-8)
day_of_week	High	+31.1% weekend vs weekday (Phase 4)
zero_streak_length / recent_nonzero_rate	High	Same dry-spell evidence as days_since_last_sale (Phase 3/9)

Medium priority: is_weekend, month, year/time trend (composition-effect caveat), rolling_std_7/28, activity_class, current_price, price_relative_to_recent_average, seasonality_cv.

Low priority / not yet recommended: week_of_year (high cardinality vs data), price_change/price_change_pct (correlational only, no promotion field), is_event_day (better used via event_name), event_type_1, event_proximity (untested).

These are CANDIDATES ONLY - the team chooses which to actually build.

14. Feature Leakage Considerations (Phase 11)

The most important section for correctness. For every candidate feature: **would we actually know this value when standing at the forecast origin, trying to predict the next 28 days?**

CENTRAL SUBTLETY: this is a fixed-origin, 28-day-ahead forecast, not a sequence of one-step-ahead predictions (unless the team chooses a recursive strategy). That changes which lag/rolling features are safe.

- lag_1/lag_7/lag_14 are only safe if computed once relative to the forecast origin and held constant across all 28 target days, or used inside a recursive forecasting loop. Recomputing them per-target-day would require already knowing near-future sales - that IS leakage.
- lag_28 (origin - 28 days) is the only single lag naturally safe for every one of the 28 horizon days.
- rolling_mean/std features must use only days up to and including the origin, then be held constant across the horizon.
- days_since_last_sale, zero_streak_length, recent_nonzero_rate, activity_class, seasonality_cv are safe ONLY if computed strictly from pre-origin history - not the full series (which would leak the target period's own behavior into its own feature).
- price_relative_to_recent_average has the same risk - the 'item's own average price' must come from a window ending at the origin.
- current_price for the horizon is likely genuinely available (prices are set in advance) but sales_long_full.parquet does not yet include those future-horizon price rows - a separate join would be needed.
- event_name, event_type, SNAP indicators, and all calendar fields are safe without qualification - the calendar is fully known in advance.
- Raw categorical 'year' is a different kind of risk: the forecast horizon contains year values with little/no training history - a plain category encoding will not generalize.

15. Potential Novelty Directions

These are hypotheses for team discussion, not a decision. Each is motivated by a specific EDA finding.

1. Pre-listing-aware zero handling

The near-perfect alignment between leading zero blocks and leading no-price periods (Section 6) suggests explicitly separating 'not yet listed' zeros from 'listed but no demand' zeros.

2. Intermittent-demand-aware modeling using dry-spell state

days_since_last_sale is the strongest single signal found (Section 12) - a novelty direction could center the approach on this state (e.g. a two-stage occurrence/magnitude structure).

3. Named-event-aware forecasting

A blunt event flag hides massive opposite effects (Christmas vs LaborDay, Section 8) - build a per-event effect model instead of one coefficient.

4. Segment-specific or hierarchical modeling

Real, data-supported heterogeneity across activity class, category, and seasonality (Section 11) supports pooling strength within segments rather than one global model.

5. Price-change as a cautious demand-shock signal

Sales rise after both price increases and decreases with a median effect of zero (Section 10) - treat detected price changes as an uncertainty/attention signal rather than a direct causal feature.

16. Key Findings for Team Discussion

20 findings identified; the most important are summarized below. Full list of 20 in EDA_REPORT.md.

#	Finding	Modeling implication
1	68.0% of rows are zero sales; skewness 17.1	Consider count/zero-inflated models or two-stage approach
2	Top 10% of series drive 54.4% of total sales	Consider volume-aware evaluation, not just overall RMSE
5	Leading zero blocks align with leading no-price periods (99.48% within 7 days)	Consider excluding/flagging pre-listing periods
7	Weekend sales +31.1% vs weekday, consistent across categories	High-priority, low-risk feature
8	Yearly growth confounded with assortment growth (active series nearly doubled)	Any trend feature needs this controlled for
10	Named events: Christmas -99.95%, LaborDay +27.5% (opposite directions)	Use event identity, not a blunt event flag
11	SNAP effect concentrates in FOODS (+17.3%, +32.3% in FOODS_2)	High-confidence, low-risk feature
14	rolling_mean_7 (r=0.820) is the strongest predictor found	Prioritize rolling-window features over single lags
15	P(sale today) falls from 65.2% to 0.6% across dry-spell buckets	days_since_last_sale should be top priority
17	Fixed-origin 28-day task: most lag/rolling features only safe relative to origin	Biggest correctness risk identified - define features relative to origin

17. Limitations

- This EDA characterizes historical, already-observed data only (d_1-d_1941). It says nothing directly about how patterns will hold in the 28-day forecast period.
- Event-lift, price-change, and per-series sensitivity analyses are all correlational, not causal.
- Per-series event/price sensitivity (Section 11) is based on a sample of 501 of 30,490 series.
- The pre-listing interpretation of leading zero blocks (Section 6) is a well-evidenced hypothesis, not a certainty - no direct listed/not-listed field exists.
- 2016 data is partial (through May 22 only).

18. What Was NOT Done

- No forecasting model was trained; no algorithm was selected; no hyperparameters were tuned.
- No final 28-day forecast was produced.
- No final ML training dataset (with engineered features) was created.
- No project novelty was chosen - Section 15 lists directions for team discussion only.
- No zero values were removed/replaced; no sales spikes were removed; no stockout/promotion labels were invented.
- raw_dataset/ and sales_long_full.parquet were not modified.

19. Recommended Next Step

Discuss the findings and candidate features in this report as a team, then jointly decide: (a) the forecasting strategy (global vs. segment-specific, recursive vs. direct multi-horizon - which directly determines which lag/rolling features are usable), (b) which candidate features to actually build, respecting the leakage constraints in Section 14, and (c) which novelty direction (if any) to pursue. Only after that should feature engineering and the final ML training dataset be built.

Final Validation

Check	Result
raw_dataset/ file checksums	Unchanged (see EDA/statistics/audit_metadata.json)
sales_long_full.parquet checksum before vs after this stage	Unchanged
Row/column shape of sales_long_full.parquet	Unchanged: 59,181,090 x 22
Zero-sales count recomputed fresh vs PROCESSING_REPORT.md	Identical: 40,241,819
Sales sum recomputed fresh vs PROCESSING_REPORT.md	Identical: 66,927,173
Every chart traces to a script-written statistics/table file	Confirmed by construction
No future sales information used in any statistic	Confirmed - restricted to d_1-d_1941